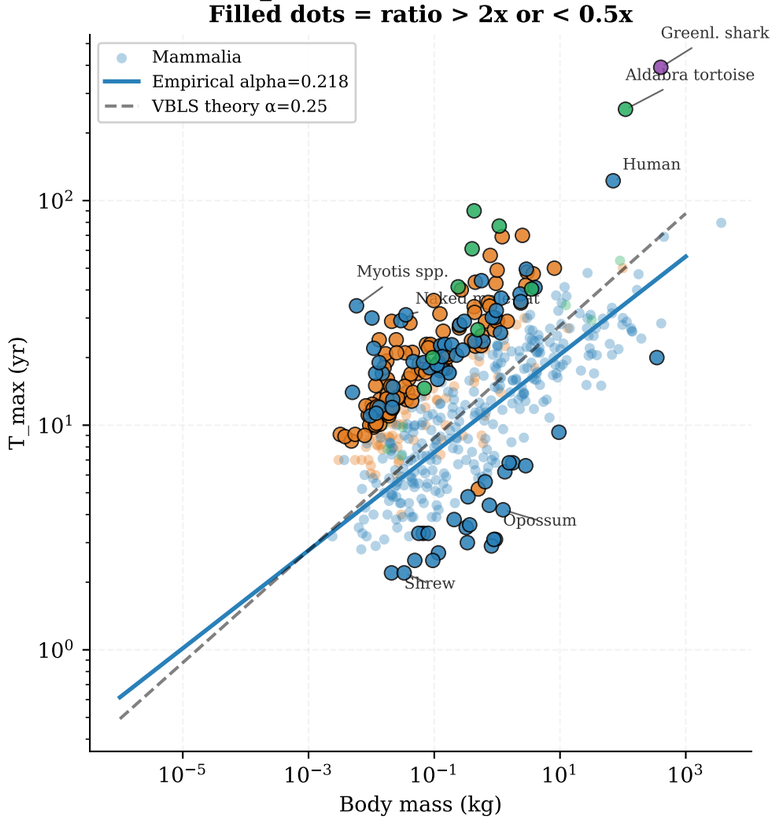
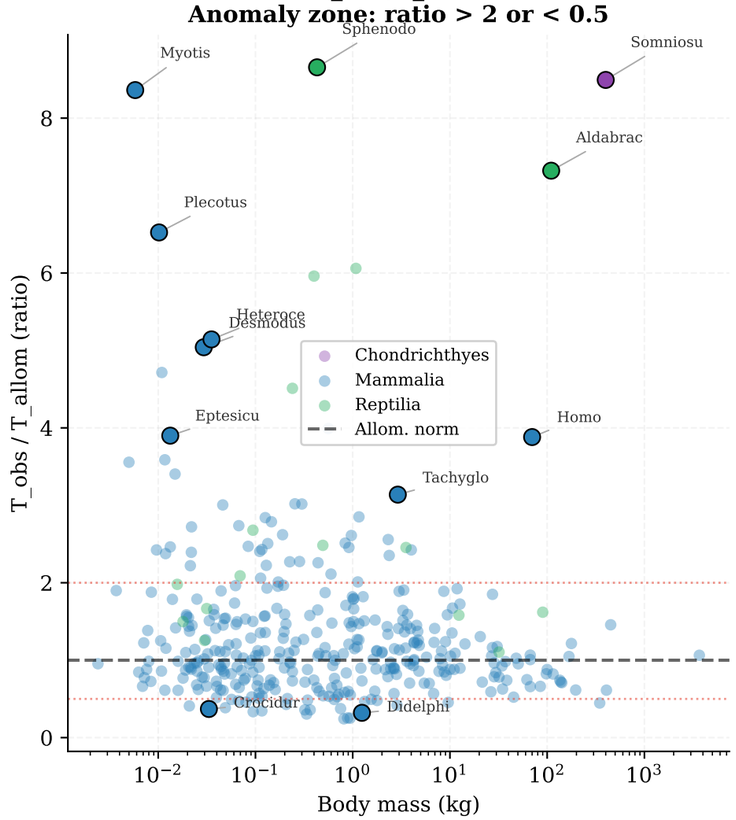
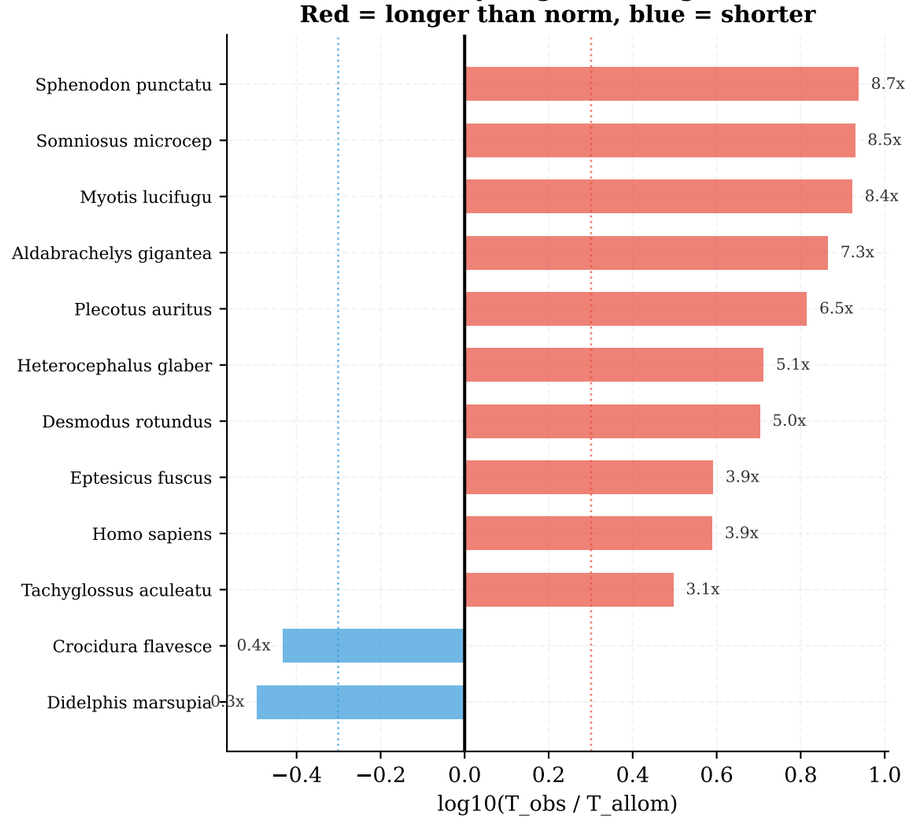
A. T_max ~ M: anomalies marked
Filled dots = ratio > 2x or < 0.5x



B. Ratio T_obs/T_allom vs mass
Anomaly zone: ratio > 2 or < 0.5



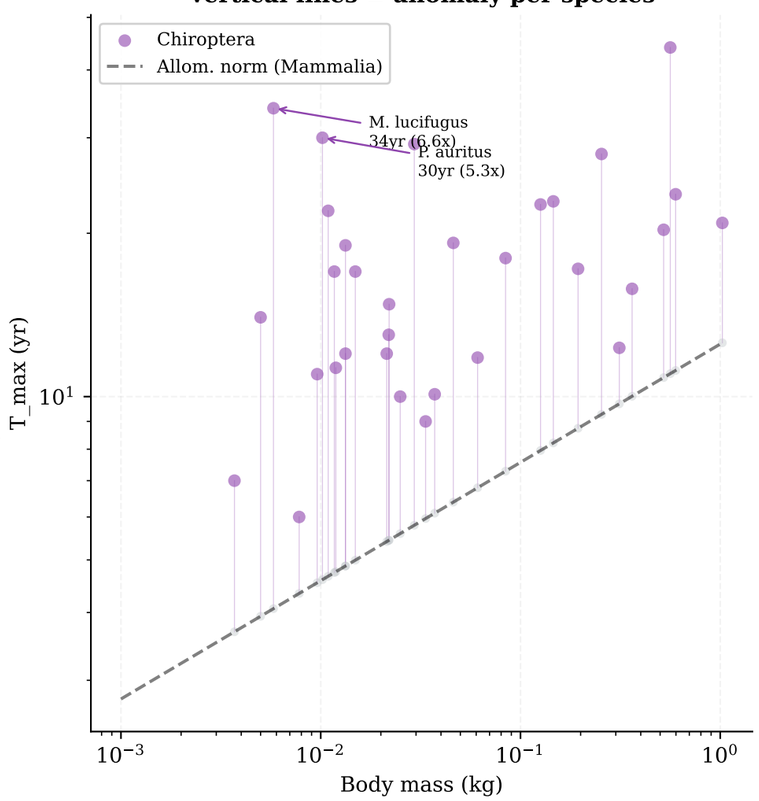
C. Anomaly magnitude (log scale)
Red = longer than norm, blue = shorter



D. Anomaly typology in terms of VBLS
Each type corresponds to one parameter of system (S)

Type	Ratio	Parameter (S)	Biological mechanism (ex
High metabolism	<1	sigma0 up	High ROS, fast DNA wear (shrews, opossums)
Strong repair (+)	>2	rho up	Highly efficient DNA repair (naked mole-rat, Homo)
Low metabolism (	>2	sigma0_eff down	Antioxidants, slow turnover (tortoises, ectotherms)
Hibernation (+)	>2	nu0 down (seas.)	Effective < chronological lifespan (bats: Myotis, Plecotus)
Polar cold (+)	>10	nu0 << mammal	T~0C: degradation x100 slower (Greenland shark)
r-strategist (-)	<0.5	rho down	Evolutionarily low repair invest. (opossums, mice)

E. Chiroptera: all species vs allometry
Vertical lines = anomaly per species



F. Summary: each anomaly -> one VBLS parameter

```
VBLS interpretation of anomalies: key conclusions

^ ANOMALOUSLY LONG-LIVED:

Chiroptera (Myotis, x6.6):
  sigma0_eff down from flight — AMPK/FOXO3 activation
  of antioxidants reduces ROS production despite
  high metabolism. The flight paradox.

Heterocephalus glaber (x4.3):
  rho up — PARP1 x3, ribosomal fidelity up,
  hypoxic adaptation -> low ROS

Homo sapiens (x4.2):
  rho_eff up via medicine and social protection;
  biological rho not unique, but cultural
  'additional rho' is species-specific

Somniosus microcephalus (x~100):
  nu0 down down down at T~0C: van't Hoff rule
  gives x2-3 per 10C -> x100+ over 40C

v ANOMALOUSLY SHORT-LIVED:

Crocidura (x0.3):
  sigma0 up — metabolism 35 W/kg, heart
  1200 bpm -> maximal ROS

Didelphis (x0.28):
  rho down — r-strategist, evolutionarily optimized
  for reproduction, not longevity

----------------------------------------
All anomalies -> one parameter of system (S).
No anomaly requires exclusion from the theory.
```